

RESEARCH ARTICLE

Utilitarian Expressions for Equilibrium Bedload Transport

Basil Gomez¹  | Philip J. Soar² ¹Department of Geography and Environment, University of Hawai'i, Mānoa, Honolulu, Hawaii, USA | ²School of the Environment and Life Sciences, University of Portsmouth, Portsmouth, UK**Correspondence:** Basil Gomez (basilg@hawaii.edu)**Received:** 22 January 2025 | **Revised:** 26 March 2025 | **Accepted:** 27 March 2025**Funding:** The authors received no specific funding for this work.**Keywords:** bedload transport formula | equilibrium bedload transport | rivers and flumes | shear stress | stream power

ABSTRACT

Heterogeneous field and laboratory measurements of equilibrium bedload transport can be collapsed to coherent relations by: expressing flow strength (x) as the ratio of the Shields number to the critical Shields number, or the ratio of dimensionless specific stream power to a reference dimensionless specific stream power; adjusting the dimensionless transport rate (y) to accommodate temporal variations in flow depth and bedload size; and treating the threshold shear stress as a variable parameter. The generalised form of the empirical functions we elaborate is: $y = c_1 x^{e_1} (1 + c_2 x^{e_2})^{e_3}$; and uncertainty surrounding estimates of y is accounted for by specifying prediction intervals for a confidence level of 90%. Their effectiveness is demonstrated by independent applications to analogous cases; and we anticipate these functions will afford a practical approach for estimating cross-section average bedload transport rates in a wide range of fluvial systems where similar conditions can be assumed to exist. Our analysis also suggests that, although it has been appreciated that data obtained in the laboratory may not be directly comparable to measurements made in the field for more than seven decades, the inveterate use of a constant value to represent the threshold shear stress has unwittingly served to obscure the disparity between rivers and flumes.

1 | Introduction

Beginning with Du Boys (1879), after nearly a century and a half of research it remains to be established that it is possible to reliably predict bedload transport rates under equilibrium conditions in natural rivers, across transport regimes that range from sporadic entrainment events on an otherwise quiescent bed in small ephemeral rivers to intense and continual motion in big perennial rivers. This is, in large part, because the prevailing bedload transport rate depends on bed state (see Gomez 2022).

Bagnold (1966) maintained that high bedload transport rates are only attainable when the mobile particles occlude the stationary surface below, and lower rates predominate when saltating and rolling particles travel intermittently over the bed. That is, he (Bagnold 1966) appraised the equilibrium bedload transport rate from the perspective of circumstances in which the transporting

flow attains a steady state and the sediment in transport has unlimited availability (see Ancey and Recking 2023). Herein, following the definition provided by Gomez and Church (1988), we assume that in such rivers the bedload and bed material are similar in character and the maximum amount of material that can be moved under prescribed, relatively constant flow conditions is in fact being moved (i.e., an equilibrium state exists in the field that is expected to be compatible with laboratory conditions).

Gomez and Soar (2022) found, by referring to measurements spanning 12 orders of magnitude obtained over the past nine decades from channels ranging in size from laboratory flumes to large sand- and gravel-bed rivers, that if the availability of transportable material is unconstrained and the bed is fully mobile, the bedload transport rate attains the expected proportional relation with flow strength. Gomez and Soar (2023) subsequently demonstrated that an elementary scale correlation between the

transport rate and flow strength provided a realistic portrayal of the range of transport rates experienced in ‘high availability’ rivers with a sporadically mobile gravel armour and a characteristic bedload size that approximates that of the substrate. However, they (Gomez and Soar 2023) also acknowledged that there were limitations to their approach because, if bedload transport is construed to occur at a negligible rate below some critical or reference value, the transport rate will not always vary in direct proportion to the chosen measure of flow strength (see Shih and Diplas 2018).

Two measures of flow strength; namely, shear stress (e.g., Du Boys 1879; Meyer-Peter and Müller 1948) and stream power (Bagnold 1960, 1966), are inveterate components of conventional thinking. The mean shear stress on the bed is habitually expressed as $\tau = \rho g Y S$: where ρ is the mass density of water; g is the gravitational acceleration; Y is the mean flow depth; and S is the energy gradient as water surface slope. Likewise, the power supply per unit channel length, $\Omega = \rho g Q S$, is the time rate conversion of water's potential energy into kinetic energy as it descends the gravity slope, and $\omega = \Omega/W$ is the mean specific stream power per unit area of the bed (Bagnold 1960, 1966): where Q is water discharge and W is the flow width. Inasmuch as $\omega = \Omega/W = \rho g Y S U = \tau U$: where U is the mean flow velocity, the two measures are closely related.

It is also appropriate to note that variations in the size and relative positioning of sediment particles on the bed and fluctuations in the flow field make it virtually impossible to define precisely the conditions under which sediment particles just begin to move (Buffington and Montgomery 1997). For this reason, there is no generally accepted methodology for quantifying the threshold or critical value of initial particle motion on a river bed (Dey and Ali 2019). Nonetheless, for more than eight decades, it has been appreciated that, by using dimensionless variables, it may be possible to collapse diverse data onto a single curve (see Einstein 1942a). Most recently, the ratio of the Shields number,

$$\theta = \tau / [g (\rho_s - \rho) D] \quad (1)$$

to the critical Shields number (θ_c) and the ratio of dimensionless specific stream power,

$$\omega^* = \omega / \rho [(\gamma_s - 1) g D]^{3/2} \quad (2)$$

to a reference dimensionless specific stream power (ω_o^* : where ρ_s is the mass density of sediment; D is a representative particle size; and γ_s is the specific gravity of sediment) have been utilised, to great effect, to collapse nondimensional measures of the transport rate obtained in both rivers and flumes to credibly coherent relations (Recking 2010; Eaton and Church 2011).

In this paper, we exploit θ/θ_c and ω^*/ω_o^* to develop new empirical relations between the relative flow strength and a dimensionless measure of the bedload transport rate, that are broadly applicable to rivers in which there is a high availability of sediment in relation to runoff and the bedload is of similar character to the bed material. Our approach relies on a compilation of diverse data obtained in rivers where the bedload

transport rate (hereafter transport rate) generally is observed to increase continuously and in proportion to some measure of the flow strength because, even for flows just above the entrainment threshold (see Cohen et al. 2010), there are no obvious constraints on the amount of potentially mobile sediment available at the bed surface (see Gomez and Soar 2023). These data encompass estimates and measurements made in intermittent, ephemeral and perennial rivers using stream-wide or limited-width traps and by at-a-point or systematic sampling. We also effect a comparison between these real-world data and disparate measurements obtained during flume experiments with uniform sediments, catalogued by Recking (2010) and, in so doing, help to resolve the decades-old debate about whether or not equations devised to fit experimental data obtained in small laboratory flumes are applicable to field situations (see Vanoni 1948; Anonymous 1954).

2 | Data Compilation and Methods

Our analysis relies on ~1400 field and ~1300 laboratory measurements of bedload transport, amassed over more than a century and gleaned from the literature (Figure 1).

2.1 | Field Data

In keeping with our previous work (Gomez 2006; Gomez and Soar 2023), we draw on measurements made in rivers in which neither the bed configuration nor stability regulate those fractions of the bed material on which transport depends, and consequently move bedload very efficiently across the entire measured range of ω^* . Reliance on ω^* (or θ) requires knowledge of the size of the sediment in motion; which, paradoxically, often is not specified. Thus, numerous field data are complemented by approximations that are dependent upon the size distribution of a textural facies in the local bed material (Gomez 2022). The latter approach does not seem unreasonable in sand-bed rivers (Bagnold 1977, 1980), but in gravel-bed rivers the underlying assumption is either that the bed is unarmoured and fully mobile (e.g., Reid and Laronne 1995); or the material in transport comprises those size fractions on the bed surface that either move with some regularity (e.g., Yuill et al. 2010), or are readily mobilised by vertical and horizontal reworking (e.g., Neill et al. 1984; Knott et al. 1987). Additionally, although transport rates are often determined when discharge is steady and the water surface slope is assumed to be constant, a second critical quantity is the flow depth (Bagnold 1977, 1980), because the transport rate increases as flow depth decreases (Williams 1970). For this reason, Bagnold (1977, 1980) argued that a correcting factor was needed to compensate for the temporal variations of depth encountered in natural rivers.

Like the vast majority of field measurements of fluvial bedload (see Hinton et al. 2017), those we rely on were acquired in single-thread channels, or (in the case of the Pasig-Potrero River) individual braids. We are aware of measurements made in 11 rivers that transport bedload very efficiently across the observed range of ω^* , for which both the size distribution of the bedload and flow depth are completely parameterised (see Supporting Information). For an additional seven rivers, the

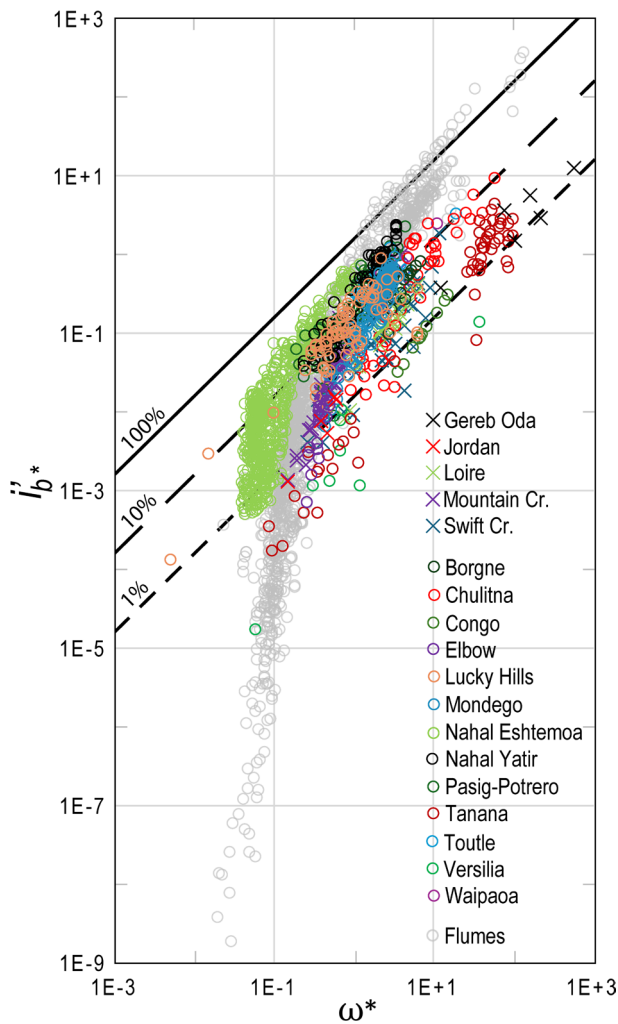


FIGURE 1 | Comparison of field and laboratory measurements of the dimensionless transport rate, corrected for variations of Y/D_{50} with flow strength, versus dimensionless specific stream power. The crosses denote sand-bed and the open circles gravel-bed rivers, and the family of diagonal lines represents specified bedload transport efficiencies (see Bagnold 1977). Measurements made in laboratory flumes and assembled by Recking (2010) are represented by grey open circles. The field data are described and all the data are catalogued in the [Supporting Information](#).

flow depth was specified but the size distribution of the bedload was assumed to be equivalent to that of the bed material (see Figure 2). Consistent with the well-established notion that if the bed is fully mobile, the size distribution of the material available for transport at the bed surface can be considered to be representative of the range of particle sizes expected to be entrained by the flow (Einstein 1944; Parker 1990), these data exemplify rivers that are also able to transport bedload very efficiently across a wide range of flows (e.g., Bagnold 1980; Reid and Laronne 1995; Cohen et al. 2010). Taken as a whole, transport rates in these 18 rivers rank amongst the highest observed in nature. However, we caution that with the exception of Nahal Eshtemoa (see Cohen et al. 2010), because most active sampling programs have limited temporal resolution (Gomez et al. 2022), the field measurements provide a window on a river's transport regime, rather than a perspective on the relation the transport rate has to mobilising flows.

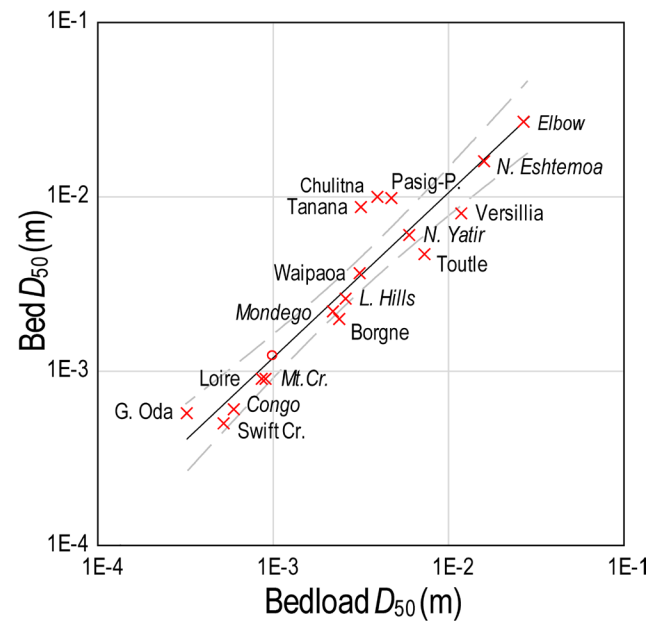


FIGURE 2 | Relation of the characteristic median particle size of the bedload (the mean value for each data set; see Gomez 2022) to the reported median particle size of the bed material ($r^2 = 0.87$). Italic type denotes rivers in which the de facto size of the bedload was assumed to be that of the bed material. The solid line represents the regression relation and dashed lines delineate the 95% confidence intervals. The East Fork River, Wyoming, (unlabeled open circle, see Figure 7) is not incorporated in the regression relation.

The mode of transport in our sample of 18 rivers ranges from incipient suspension to bedload, moving as multi-scale sand and low-angle gravel bedforms, or sheets and individual particles and, with two exceptions, the bed is unarmoured. In the armoured Elbow River, the active bed is about one-third of the channel width at the measuring site and the sediment in transport originates from bar deposits upstream, that are remobilised during the spring snowmelt (Hudson 1983), and the estimates for the Jordan River pertain to coarse sediment transport during and after a catastrophic flood that completely reshaped the channel (Inbar and Schick 1979). Within a given river, the transport rate was determined using a single apparatus or technique, and the data encompass measurements obtained by at-a-point sampling near the mouth of one of Earth's largest perennial rivers to those acquired by continuous sampling in small, low-order, ephemeral, headwater channels. Except for Lucky Hills, where the transport mode of a grain-size fraction was determined by its Rouse number (Yuill 2009), the proportion of the fluvial sediment load described as bedload and the transport rate are what were measured directly or estimated indirectly using a given device or methodology (i.e., using stream-wide or limited-width traps, by at-a-point or systematic sampling, or by estimating the volumetric change in sediment deposits created by exceptional floods in which most bed material sizes were entrained). Twelve (one intermittent, four ephemeral and seven perennial) rivers possess natural bedload transport regimes, but those of the five remaining perennial rivers have been perturbed by a natural or anthropogenic disturbance; and a second intermittent river was subject to periodic flushing flows. Collectively, these measurements yield 1390 values (see [Supporting Information](#)).

It is a prerequisite of most laboratory experiments that all particle sizes present on the bed are in motion and, to ensure compatibility between laboratory and field conditions, a near equivalence between the calibre of the bedload and a representative (typically solitary) bulk or subsurface sample of the bed material has been invoked to identify those field measurements that represent equilibrium transport (see Gomez and Church 1988). The Jordan River aside, where the size of the bedload and bed material are not definitively known (Inbar and Schick 1979), and accepting that the disparity will be more pronounced if, for example, not all the sediment in motion originates from the bed in the immediate vicinity of a sampling location (e.g., Neill et al. 1984; Knott et al. 1987; Hayes et al. 2002), or the size distribution of the bedload varies as gravel bedforms migrate and/or adjust to the prevailing flow conditions (e.g., Dinehart 1992), we find that there is a near 1:1 correlation in the relation between the characteristic, median size of the bedload and the reported median size of the bed material (Figure 2). On this basis, we construe that the field data we utilise generally reflect conditions that are broadly compatible with, but not necessarily equivalent to, those experienced in laboratory flumes.

2.2 | Laboratory Experiments With Uniform Sediments

Recking (2006, 2010) used rigorous selection criteria, explicitly formulated to ensure that the preferred experimental conditions reflected those experienced in gravel-bed rivers with either partially (at or near the threshold of particle motion) or fully mobile beds, to compile a data set from the literature. Recking (2006) applied Vanoni and Brooks' (1957) side-wall correction to these data; which encompass the classic experiments of Gilbert (1914), Paintal's (1971) investigations into the condition at which bedload transport is initiated, and 10 other laboratory studies of bedload movement. The historical records were amalgamated with newly acquired data, and the entire catalogue appertains to steady-state, uniform flow conditions (Recking 2010); although, as is the case for the real-world data, it encompasses measurements involving sand-sized bed materials and high relative depths. Scrutiny of the 1317 measurements catalogued by Recking (2010) revealed 14 cases where apparent inconsistencies could not be resolved with the information available to us. These data are not included in our analysis, which consequently relies on 1303 values (see Supporting Information).

2.3 | Parameterising θ_c , ω_o^* and the Transport Rate

The vast majority of the data we draw on comprise observations of the fundamental quantities of water discharge, the flow depth, the flow (channel) width and the transport rate, and some measure of slope and the size of the material (actually or presumed to be) in motion. Additional parameters, which usually were not specified in the field, including the mass densities of water and sediment, were assumed to be constant ($\rho = 1000 \text{ kg/m}^3$ and $\rho_s = 2650 \text{ kg/m}^3$). To compute θ and ω^* , we assume $D = D_{50}$, $\tau = \rho g Y S$ and $\omega = \rho g Q S/W$; thus, by definition, the mean flow

velocity, $U = \omega/\tau$: where D_{50} is the particle size for which 50% of the bedload (or bed material) is finer by weight.

Like Bagnold (1977), we assume that a residual stream power, or shear stress, is required to maintain transport (the condition we seek to characterise); but, although the point at which motion is initiated is a fundamental quantity for sediment transport, there is no definitive value or mode of computation for either θ_c or ω_o^* , which are multiparameter functions (Buffington and Montgomery 1997; Ferguson 2005). For practical application, the empirical Shields curve is a firmly established methodology for specifying the condition of incipient motion of a representative particle size (Buffington and Montgomery 1997); to which Soulsby and Whitehouse (1997) fitted a modified, and now widely utilised (e.g., USACE 2023), relation of the form:

$$\theta_c = 0.3 / (1 + 1.2 D^*) + 0.055 (1 - \exp[-0.02 D^*]) \quad (3)$$

where $D^* = D ([g \{ \gamma_s - 1 \}] / \nu^2)^{1/3}$; and ν is the kinematic viscosity of the water (which we assume to be $1.0034 \text{E-}06 \text{ m}^2/\text{s}$, throughout). Representing Shields' (or any other empirical) threshold curve as a single line is a deterministic solution to an inherently stochastic phenomenon that lacks a standard definition and is inhibited or enhanced by a myriad of factors, but Paphitis (2001) observed that the most commonly used formulations provide comparable threshold determinations. That said, we also accept that the conditions governing the initial movement of discrete particles under controlled (laboratory) conditions may differ from those that determine the equilibrium transport rate in nature, and we found that applying this function of D^* to all the laboratory and field data yielded inconsistent results (i.e., there were cases when no transport was predicted) in 4 perennial and 2 ephemeral rivers. We attribute this to the fact that the shear stress required to initiate motion is greater than that needed to maintain transport. Additionally, θ_c is known to exhibit values ≈ 0.01 when particles are exposed on, rather than hidden in, the bed (Fenton and Abbott 1977), or are already in motion (Montgomery et al. 1999). For these reasons, we applied a constant value of $\theta_c = 0.01$ to these 6 gravel-bed rivers (see Supporting Information).

Einstein (1944) suggested that the transport rate depends on the bed composition and flow resistance, and Eaton and Church (2011) related ω_o^* to θ_c and flow resistance, such that:

$$\omega_o^* = \theta_c^{3/2} / (U / u^*) \quad (4)$$

where the shear velocity, $u^* = \sqrt{(g Y S)}$. Accepting that, as noted previously, θ_c tends towards a constant value in some rivers, we used Eaton and Church's (2011) shear stress-based Equation, that preserves a dependence on flow resistance, to compute ω_o^* throughout.

We continue to employ a dimensionless form of the transport rate that is analogous to the dimensionless parameter (Φ) employed by Einstein (1942a)

$$i_{b*} = i_b g / \rho [(\gamma_s - 1) g D_{50}]^{3/2} \quad (5)$$

to relate sediment transport to the ratios of θ/θ_c and ω^*/ω_o^* ; where i_b is the bedload transport rate expressed as immersed mass per unit width (see Gomez and Soar 2022). However, given the diversity of the data we rely on and the knowledge that some residual scatter may be abstracted by adjusting the transport rate to allow for variations of Y/D_{50} with flow strength (Bagnold 1977, 1980), we apply the correcting multiplier,

$$([Y/D_{50}]/[\bar{Y}/\bar{D}_{50}])^{2/3} \quad (6)$$

to i_{b*} where: $(\bar{Y}/\bar{D}_{50})$ is the mean ratio for each river or ensemble of flume experiments conducted at a given width; and the corrected value is designated i'_{b*} .

3 | Relations Between the Transport Rate and Flow Strength

Remarking on the coherence of the laboratory data, Recking (2010) noted that expressing θ_c as a function of S yielded uneven results across the observed range of transport rates. As Bagnold (1977) appreciated, some scatter can be eliminated by adjusting the transport rate to a common Y/D , and specifying

θ_c as a function of the bedload size also contributes to a more consistent outcome. Thus, we find that our formulations of the dimensionless ratios θ/θ_c and ω^*/ω_o^* collapse the field data onto single curves that mirror, but do not duplicate the trend exhibited by the laboratory data (Figure 3); albeit that the reduction in scatter is assuaged by the less controlled conditions and the greater uncertainty associated with measurements made in the field. That said, there are intrinsic reasons why the field data incorporated in Figure 3 plot 'lower' than the flume data.

First, we reiterate that previous attempts to collapse field and laboratory data to a single relation employ a unique value of θ_c (e.g., Recking 2010; Eaton and Church 2011). There is overlap in the data we utilise and Gomez and Church's (1988) catalogue, that Eaton and Church (2011) rely on (and we exploit Recking's (2010) compilation in its entirety), but it is our use of Equation (3) to approximate the true value of θ_c that separates the real world from the laboratory. This is because the transport rate increases as flow depth decreases (Williams 1970; Bagnold 1977), so shallow depths in laboratory channels enhance the transport rate with respect to conditions observed in many rivers. We illustrate this point in Figure 4A, by comparing the increase in $\theta-\theta_c$ needed to maintain a constant transport rate as Y/D_{50} is increased, using measurements

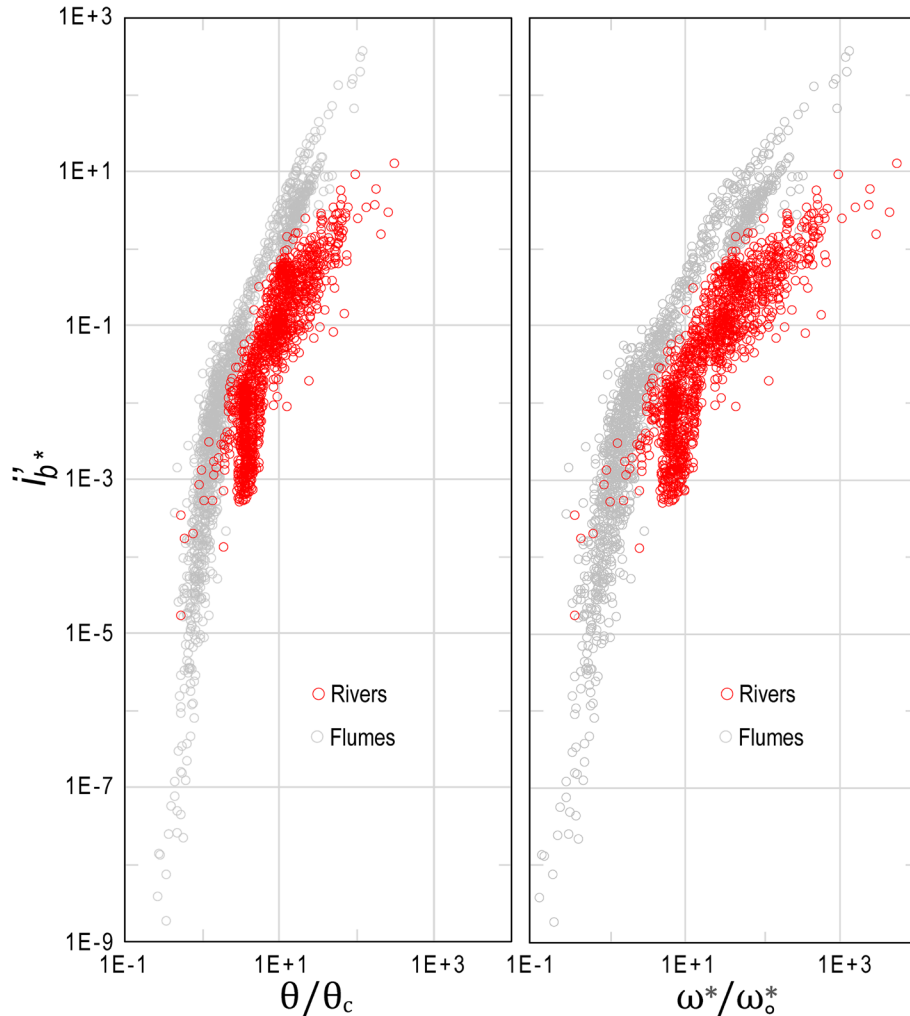


FIGURE 3 | Comparison of field and laboratory measurements of the dimensionless transport rate, corrected for variations of Y/D_{50} with flow strength, versus θ/θ_c and ω^*/ω_o^* .

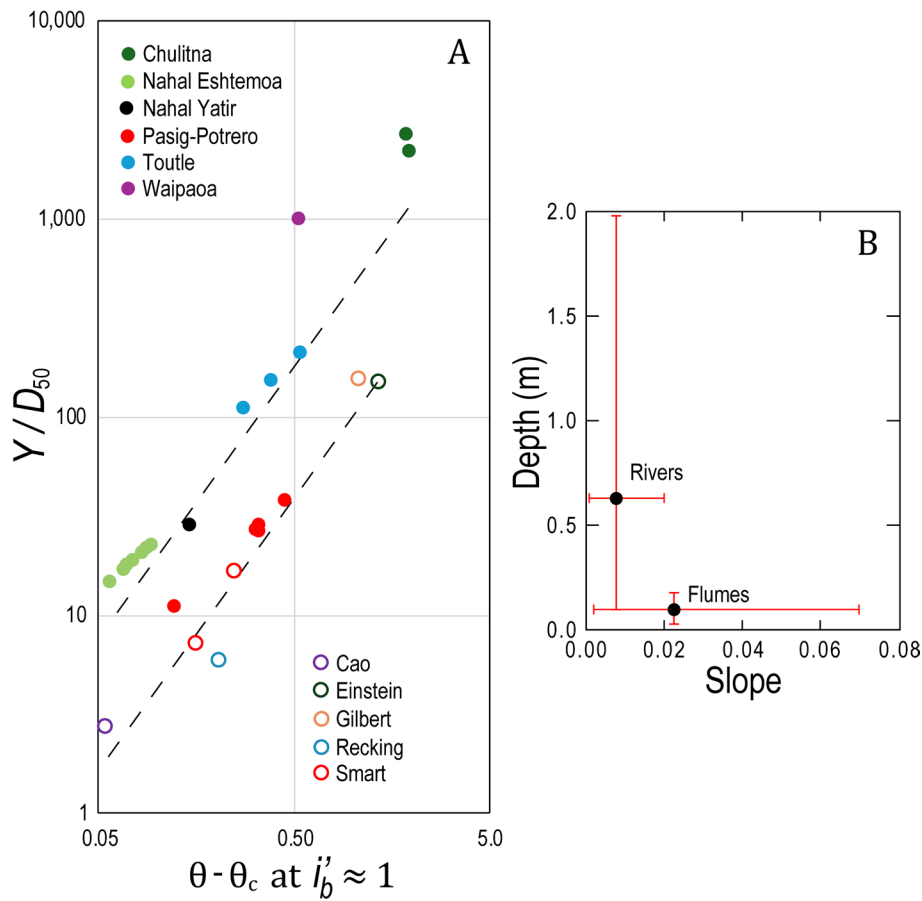


FIGURE 4 | (A) Increase in $\theta - \theta_c$ needed to maintain a constant dimensional, depth-corrected transport rate (kg/ms) as Y/D_{50} is increased in river and flumes. The regression relations (dashed lines) for the field and laboratory data are $y = 461.45 x^{1.36}$ ($r^2 = 0.93$) and $y = 101.85 x^{1.37}$ ($r^2 = 0.95$), respectively. See text for discussion. (B) Range of slope and depth in the laboratory and field data utilised in this paper. Error bars encompass the 10th–90th percentiles and solid black circles denote mean values.

made in five laboratory channels and six rivers. For the prescribed value of the dimensional, depth-corrected transport rate, $i_b' \approx 1$ kg/ms, the flumes (in which the depth and median particle size in motion are between 0.03 and 0.13 m and 0.5 and 44 mm) represent scaled versions of the natural rivers (in which the depth and median particle size in motion are between 0.1 and 3.2 m and 0.9 and 16 mm), and only in the steep, shallow, highly charged Pasig-Potrero River do the conditions in individual braids approach those simulated by some flume experiments (e.g., Smart 1984).

Shallow flow depths also give rise to lower bed shear stresses which, if the slope in a laboratory flume corresponded with that observed in nature, would be far below the critical value required to initiate motion; because the Shields' criterion for the onset of particle motion increases as the ratio of water depth to grain size decreases. To compensate for the reduction in mobility, the slope adjusts so the laboratory flow eventually transports sediment at the rate it is supplied or recirculated; thus, in general, the slope attained during flume experiments is steeper than that observed in rivers. Figure 4B illustrates the disparity between the slope and depth encountered in the laboratory and field data we utilise.

Secondly, steady flow and transport permit very high transport rates to be maintained indefinitely on steep slopes in

laboratory flumes, where the moderating effects of flow depth and resistance differ from those experienced at lower gradients (Smart 1984). Whereas in nature, although extreme events may be able to mobilise normally static bed elements and access the sediment stored within them (Warburton 1992; Lenzi 2001), an episodic sediment supply, thin alluvial cover, and high percentage of exposed bedrock ordinarily place limits on the amount of sediment that is available for transport in steep headwater channels (Montgomery and Buffington 1997). For this reason, and because the ability of a river to transport bedload at or near its theoretical capacity hinges on sediment supply (Gilbert 1876; Gomez 2006, 2022), high transport rates are observed in a much wider range of rivers with more moderate gradients, in which there is a persistently high availability of sediment in relation to runoff (Gomez and Soar 2022).

Thirdly, particle interactions and relative size effects affect the initiation of motion and transport rate of sediment mixtures, which differ from that of uniform material (Proffitt and Sutherland 1983; Samaga et al. 1986; Wilcock and Southard 1989; Kuhnle 1993); and, although the probability of movement becomes very small, it never equals zero and a low sediment discharge that is many orders of magnitude less than the threshold condition can be measured in laboratory channels with a uniform bed material, if the transport rate is determined over a protracted period of time (see Paintal 1971). Coincident with a shift

in the size range of the sediment in motion, very low transport rates are observed in some rivers with a strongly bimodal bed material size distribution (see Bagnold 1980). However, in most rivers, the degree of exposure of individual grains differs from that of particles on a nominally flat, flume bed and, even during prolonged, low-magnitude transporting events, the entrainment threshold is determined by the particle-to-particle interactions that moderate bed material shear strength (Milhous 1973; Cohen et al. 2010).

Additionally, accepting $\tau = \rho C_d U^2$ (where C_d is the drag coefficient) and $\omega = \tau U$ (so $\omega \propto U^3$, $\omega \propto \tau^{1.5}$ and $\tau \propto \omega^{2/3}$), we submit that there is an inevitable tendency for a relation involving ω to appear less coherent than one that relies on τ (see Figure 3). Whether or not this matters likely depends on how the controls on bedload transport are perceived to operate (see Ancey and Recking 2023). Viewed as a local process, the transport rate is assumed to be a function of the flow conditions (e.g., the bed shear stress distribution) averaged across the river width. Whereas non-local transport implies that the measured rate at a specific location is dependent on the conditions in the surrounding area, as opposed to those prevailing at a specific location. Bagnold (1966, 1977) recognised that both i_b and ω can vary substantially over short distances, and he (Bagnold 1977) specified that the general relation of i_b to ω could only be elucidated using parameter values that extend across a spatially averaged range of flow conditions and covered an appropriate span of elapsed time. The implication being that (at least, under equilibrium conditions) bedload transport is, in essence, a non-local process (see Ancey and Recking 2023). Eaton and Church (2011) also noted that the transport rate at a given cross-section inevitably depends on the ability of the flow to entrain sediment from the bed in the immediate vicinity as well as the supply of particles that are already in motion arriving from upstream; and the impracticability of obtaining stream-wide measures of S , Y and U suggests the use of mean or total hydraulic measures is unavoidable (Gomez and Church 1989). However, it remains to be determined why the threshold of motion or transport rate should be better predicted by τ than ω (Carson and Griffiths 1987; Ancey and Recking 2023); and accepting that the channel-wide integration of hydraulic values prior to the transport calculation is the rule, rather than the exception, we develop empirical expressions that utilise both τ and ω .

At conditions close to the threshold for transport, bed material releases are sporadic and the transport rate increases rapidly as the relative flow strength increases (see Cohen et al. 2010; Shih and Diplas 2018). Above the entrainment threshold, the transport rate systematically increases and eventually exhibits a proportionality with relative flow strength. Thus, the overall relation is non-linear (Figure 3). Various empirical functions have been fitted to diverse field and laboratory data, but there is no consensus about how to infer the transport rates' observed dependence on relative flow strength from data (see Recking 2010; Eaton and Church 2011; Shih and Diplas 2018; Ancey and Recking 2023). Our approach is to use a continuous hinge function to produce a single, smooth curve (and single Equation) by flexibly connecting two lines in log space (Gelman 2017; Figure 5).

The hinge function comprises three integrated relations: (i) a steep lower curve, which describes the non-linear increase in

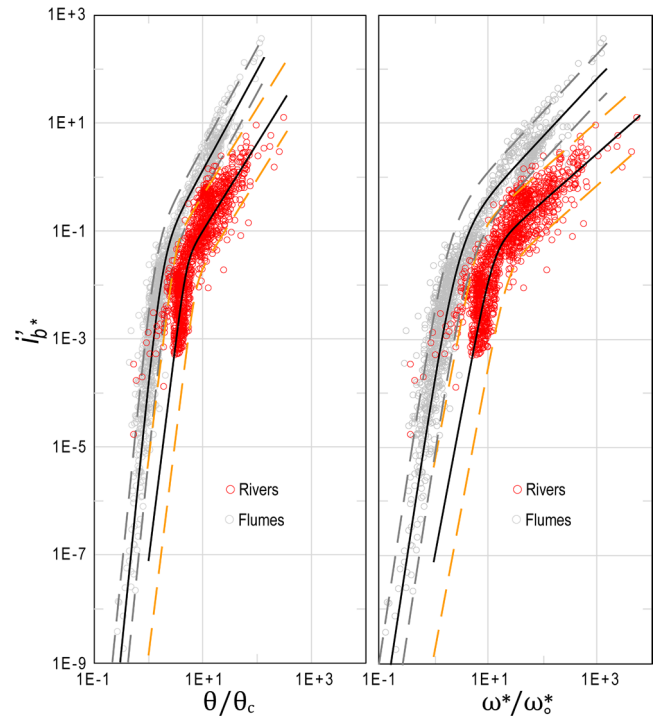


FIGURE 5 | Continuous hinge functions (solid lines) fitted to field and laboratory measurements of the dimensionless transport rate, corrected for variations of Y/D_{50} with flow strength, expressed as θ/θ_c or ω^*/ω_0^* . Dashed lines delineate the prediction limits for a confidence level of 90% (see text for discussion).

transport rate with flow strength in the vicinity of the entrainment threshold; (ii) a gentle upper curve, which describes the non-linear increase in transport rate with flow strength above the entrainment threshold; and (iii) a flexible pivot, or hinge, to model the intervening fuzzy region of flow strength and greater uncertainty, and connect the lower and upper curves. A user-specified upper value of the dimensionless sediment transport rate and lower value of the dimensionless flow strength term(s) impose bounds on the data domains associated with the lower and upper curves, which are bivariate, best-fit power functions of the form commonly applied in sediment transport modelling (see Ancey and Recking 2023). The variable gradient across the pivot region equates to the inverse logistic function, which ensures there is a smooth transition between the lower and upper curves (re-specified in log space; after Gelman 2017). Furthermore, the distance scale of the inverse logistic function (in abscissa units) affords a smoothing factor, which enables the pivot to relax from a sharp corner (the intersecting point of the lower and upper curves) to a smooth curve that best fits the data and asymptotically approaches the gradients of the lower and upper curves. The smoothing factor can either be specified manually or computationally optimised, as is our approach, to minimise the prediction error.

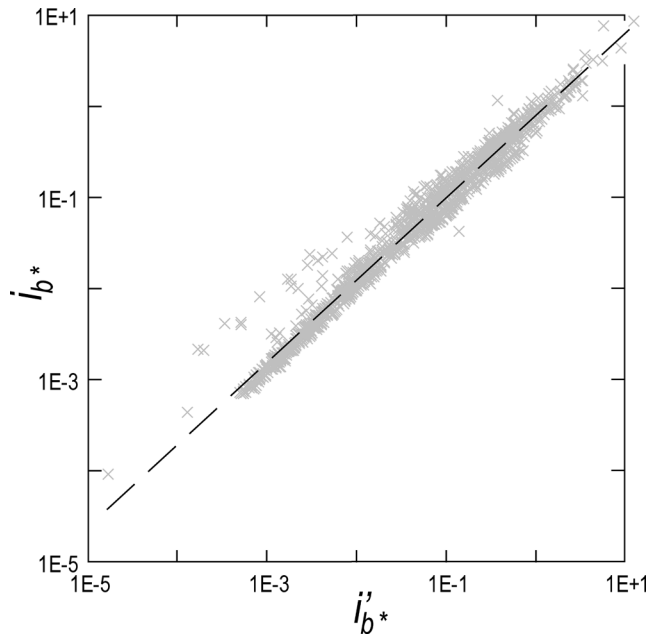
For the general case, we write:

$$i'_{b*} = c_1 x^{e_1} (1 + c_2 x^{e_2})^{e_3} \quad (7)$$

where $x = \theta/\theta_c$ or ω^*/ω_0^* ; and context-specific values for the coefficients and exponents are given in Table 1. To parameterise

TABLE 1 | Context-specific values of the coefficients and exponents in Equation (7).

		θ/θ_c			ω^* / ω_o^*		
		Fitted relation	Prediction limits		Fitted relation	Prediction limits	
			Lower	Upper		Lower	Upper
Field	c_1	9.741E-08	1.380E-09	4.375E-06	9.741E-08	1.380E-09	4.375E-06
	c_2	1.276E-05	6.734E-07	1.688E-04	1.095E-04	1.053E-05	8.655E-04
	e_1	8.083	8.083	8.083	5.389	5.389	5.389
	e_2	7.143	7.143	7.143	3.704	3.704	3.704
	e_3	−0.904	−0.904	−0.904	−1.214	−1.214	−1.214
Laboratory	c_1	2.063E-04	6.233E-06	4.973E-03	2.063E-04	6.233E-06	4.973E-03
	c_2	0.111	0.035	0.320	0.122	0.046	0.301
	e_1	10.057	10.057	10.057	6.705	6.705	6.705
	e_2	3.846	3.846	3.846	2.273	2.273	2.273
	e_3	−2.142	−2.142	−2.142	−2.459	−2.459	−2.459

**FIGURE 6** | Relation of the dimensionless transport rate, corrected for variations of Y/D_{50} with flow strength, to the unadjusted dimensionless transport rate in rivers.

Equation (7), the steep lower and gentler upper curves were arbitrarily derived for $i'_{b*} \leq 0.01$ and $\theta/\theta_c \geq 5$ and $\omega^*/\omega_o^* \geq 15$, respectively, and these bounds apply to both the laboratory and field data. Alternative values will change the nature of Equation (7) but visually, at least, those given in Table 1 provide a good description of the data (Figure 5).

In the case of the field data, we also find there is a coherent ($r^2 = 0.92$) relation between i'_{b*} and i_{b*} (Figure 6), that can be utilised to obtain the unadjusted dimensionless transport rate, such that:

$$i_{b*} = 0.799 i_{b*}'^{0.906} \quad (8a)$$

For reference, the equivalent expression ($r^2 = 0.99$) for the laboratory data is:

$$i_{b*} = 0.986 i_{b*}'^{0.966} \quad (8b)$$

The prediction limits, specified in Table 1 and delineated in Figure 5, refer to the 90% uncertainty range associated with a single response within the sample domains of the lower and upper curves, while holding the smoothing factor constant, and provide instructive bounds to the models' performance with respect to each data set. Inasmuch as they serve to illustrate and quantify the variability inherent in the data, they act to convey the predictive power of our approach. We emphasise this point, by expressing the proportion of the variance in the observed i'_{b*} values that is explained by the predicted i'_{b*} values (see Piñeiro et al. 2008), using an r^2 goodness of fit measure to summarise the predictive capability of Equation (7). Regardless of whether $x = \theta/\theta_c$ or ω^*/ω_o^* , the r^2 is 0.91 and 0.67 for the laboratory and field data, respectively; the more moderate value being a product of the continually varying conditions encountered outside the laboratory and the greater uncertainty associated with measurements made in the field. Moreover, accepting that the coefficients and exponents in Equation (7) inevitably reflect how the individual terms are parameterised, the robustness of our model is evidenced by the fact that these r^2 values remain unchanged if the critical Shields number (Equation (3)) is increased or decreased by 10%. As in our previous work (Gomez and Soar 2023), we employ bootstrap confidence limits (10000 resampled datasets) to capture the uncertainty associated with our model's explanatory capacity and corroborate its reliability. On this basis, when $x = \theta/\theta_c$ the 90% limits on r^2 ranged between 0.90 and 0.92 for the laboratory data and between 0.55 and 0.72 for the field data. The equivalent ranges for ω^*/ω_o^* being 0.90 to 0.92 and 0.56 to 0.73 for the laboratory and field data, respectively.

The quintessential values necessary to apply our formulation(s) to any high availability river are time-averaged values of: Q ; S ; W ; Y ; and the size (D_{50}) of the material in motion. The former

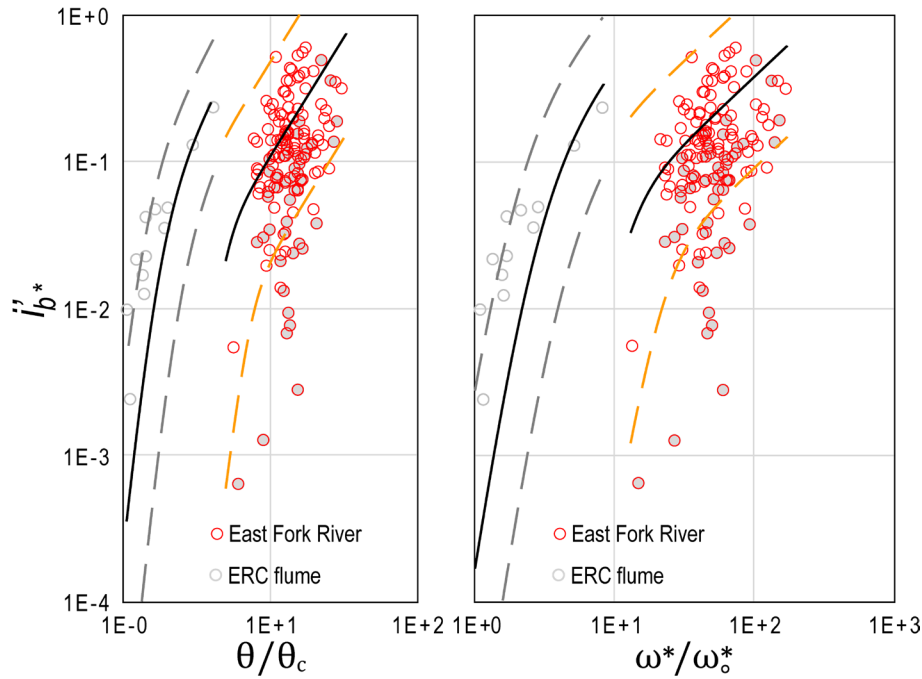


FIGURE 7 | Stream-wide transport rates in the East Fork River, Wyoming, Leopold and Emmett (1997), and the Environmental Research Centre (ERC), University of Tsukuba, Japan, 4-m flume (Ikeda 1983) compared with the trend (solid lines) and limits (dashed lines) predicted by Equation (7) (and applying the context-specific values of the coefficients and exponents given in Table 1). The shaded circles denote measurements made in the East Fork River on the waning limb of the annual snowmelt hydrograph (see text for discussion and Supporting Information).

(Q) is customarily acquired; S is often approximated by a unique value; W and Y can be extracted from rating relations; and determining D_{50} does not appear to be an insurmountable proposition in rivers that have a high availability of sediment in relation to runoff if, as a first order approximation, it is accepted that the size of the sediment in motion is indeed equivalent to that of the bed material (see Figure 2). Like other functional models (see Gomez and Church 1989), Equation (7) applies to particular circumstances and specifically to rivers in which sediment supply and availability do not impose constraints on the transport rate, which can seemingly be directly related to tangible flow and sediment parameters. Such conditions are known to exist in ephemeral dryland rivers (e.g., Reid and Laronne 1995), to prevail in overloaded perennial rivers (e.g., Simon 1999), and may be a transient response to extreme rainfall-runoff events in high gradient, montane rivers (e.g., Picco et al. 2012). Predicting transport rates in such settings is not only a scientific challenge, but also an essential precursor to ameliorating the pervasive environmental and economic costs fluvial bedload has the potential to generate (e.g., Badoux et al. 2014; Urich et al. 2021).

We offer an independent test of our empirical relation(s) by comparing their predictive ability to the stream-wide measurements obtained in the East Fork River, Wyoming (Leopold and Emmett 1997). These unique data have proved challenging to utilise, not least because coarse sand and fine gravel move as dunes and bedload sheets, from one storage area into the intervening reach on the waxing limb of the annual snowmelt hydrograph and on to the next storage area downstream on the waning limb (Leopold and Emmett 1976; Meade 1985); a pattern of motion that gives rise to clockwise hysteresis in the transport rate–flow relation. Consequently, higher transport rates were, in general, observed before, and lower transport rates recorded

after the snowmelt peak in the reach where the bedload trap was located (Leopold and Emmett 1997; Figure 7). Bagnold (1977), Gomez and Church (1989) and Gomez and Soar (2022) have drawn attention to the nonlinear temporal structure of the transport regime in the East Fork River, that is not a product of differences in the size of the material in motion (which is equivalent to that of the bed material, see Figure 2); but which, discounting temporal variations in the amount of sediment available for transport, our formulations accommodate. We also note, in passing, that our formulation(s) for laboratory experiments conducted with uniform bed material embrace the trend in the measurements Ikeda (1983) made during his investigations of low relief bedform development (including bars, high-angle dunes and antidunes) in the large, University of Tsukuba, Environmental Research Center (ERC) flume (Figure 7).

4 | Predicting the Magnitude of Bedload Transport

Although it has long been accepted that there may be no basis for judging if formulas based on laboratory data are applicable to natural rivers (Vanoni et al. 1961), many commonly used bedload transport formulae are based entirely on laboratory data (e.g., Meyer-Peter and Müller 1948; Wilcock and Crowe 2003); supplemented with or supported by measurements made in sand-bed rivers (e.g., Einstein 1950), or rivers that transport mixtures of sediment sizes (Wilcock and DeTemple 2005). At first glance, a foundation laid on laboratory data appears sound, because both field and flume data seemingly overlap (see Figure 1; Gomez and Soar 2022), as is the case when $D = D_{84}$ and θ_{84} is scaled with $\theta_c = \theta_{c84} S$ (Recking 2010); and, as previously noted, they collapse to a single relation if a unique value for θ_c is used when calculating ω_c^* (Eaton and Church 2011). This concordance is not

preserved in our analysis (Figure 3), which reveals that, for the case of equilibrium bedload transport, there are important and readily explainable differences in the trends seen in data derived from experimental studies and measurements made in natural rivers.

We emphasise, like Bagnold (1977), that although the representative size utilised refers to the sampled bedload, not the bed material, we maintain it is appropriate to revisit the long-standing and, to date, unresolved issue of whether bedload formulae devised to fit measurements made during controlled laboratory studies, such as the Meyer-Peter and Müller (1948) formula, are applicable to natural channels (Vanoni 1948; Anonymous 1954; Vanoni et al. 1961). This is because the disparity we bring to light may help to explain why empirical formulae, like that of Meyer-Peter and Müller (1948), overpredict the observed transport rates attained in high availability rivers (see Gomez 2006). To illustrate this point, and recalling that $i_{b*} \approx \Phi$, in Figure 8 we portray the transport relation expressed by Wong and Parker's (2006) modified MPM formula, which was presented in the dimensionless form suggested by Chien (1954):

$$\Phi = 3.97 (\theta - 0.0495)^{1.5} \quad (9)$$

Data from Gilbert's (1914) and Meyer-Peter and Müller's own experiments (some of which are incorporated in the laboratory data we utilise) were used to derive the original empirical relation that incorporates effects due to variations in sediment density and resistance (Meyer-Peter and Müller 1948). However, Wong and Parker's (2006) reanalysis revealed that, in the case of lower-regime, plane-bed, equilibrium transport of uniform sediment, the bed roughness correction is unnecessary. The modified relation replicates the trend of the laboratory data, but we contend that it will inevitably perform poorly in field situations because transport rates measured in laboratory flumes are not directly comparable to those experienced in nature (see Bagnold 1977).

It is also apposite to note that the methodology for evaluating the performance of a formula, wherein computed and observed values or relations are compared, which has remained essentially unchanged for more than eight decades (Johnson 1939; Einstein 1942a; Einstein 1942b), is not the same as that needed to estimate the sediment load. In the former case, numerous studies have demonstrated that most formulae yield neither an approximate 1:1 correlation nor an acceptable statistical comparison (see Gomez and Church 1989), and their generally poor performance has stimulated efforts to improve their predictive accuracy. One such approach has been to try to reduce the uncertainties associated with key input parameters, such as shear stress and grain size (e.g., Monsalve et al. 2016). Another has been to develop site-calibrated relations (e.g., Wilcock 2001; Bakke et al. 2017). Our approach is to continue to use primitive, readily attained, site-specific information to assess how a formula is able to replicate observed states and, because there is no a priori requirement for a transport rate to be uniquely dependent on the flow conditions and the sources of variability associated with samples and measurements made, field data are unknown, we have specified prediction intervals for the 90% uncertainty range (Table 1).

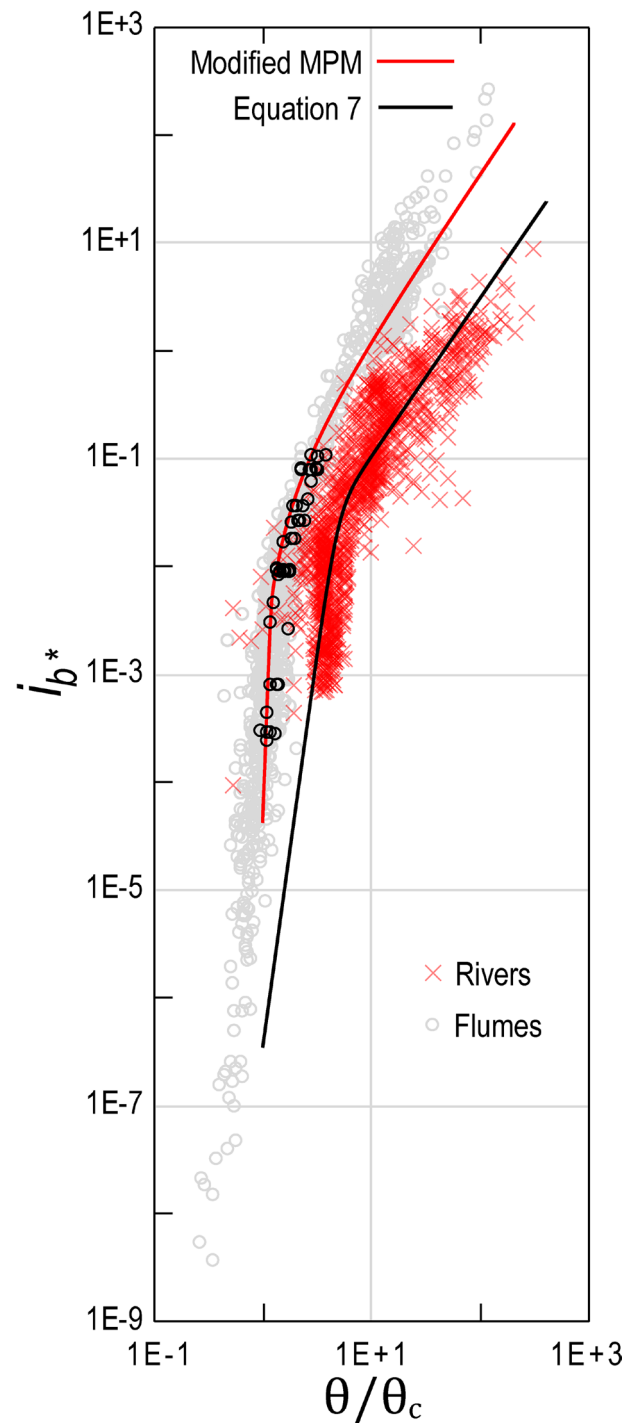


FIGURE 8 | Wong and Parker's (2006) modified Meyer-Peter and Müller (MPM) bedload transport relation (solid red line). The black open circles denote laboratory data Meyer-Peter and Müller (1948) used to derive their original empirical relation. The empirical relation developed in this paper (Equation (7), solid black line) is shown for comparison. See text for discussion.

Assuming that the transport rate in a river can be related to discharge, or some other closely related measure of the driving force, and the sequence of discharges is unimportant, Einstein (1944) demonstrated that it is possible to determine the annual amount of sediment transported through a reference section from the flow duration curve. We believe our formulations now present the opportunity to do this in a wide range of rivers, that are

representative of a corner of the universe of fluvial systems in which an equilibrium state prevails and exhibit transport rates at the high end of the reported range in nature. Consequently, the relations, between i'_{b*} and ∂/∂_c or ω^*/ω_o^* we have developed, delineate envelope curves for the maximum transport rate associated with the given measure of flow strength (in both rivers and flumes). Accepting that bedload transport is amenable to an empirical treatment without direct observation, the trend in the field measurements we rely on potentially affords the opportunity to assess how, for example, the transport rate responds if the constraints imposed by sediment supply and availability are removed or there is a change in hydrologic regime. Such conditions are expected to be increasingly experienced in sub-humid regions, as the anticipated climate change towards increased aridity starts to influence the amount of sediment entering and moving as bedload in tropical and temperate river systems (see Armoza-Zvuloni et al. 2022); and in glacier-fed rivers, in which it is thought that, as the climate warms, the flows on which bedload transport depends will transition from (supply-limited) meltwater- to (transport-limited) precipitation-driven events (see Lane et al. 2021). Our formulations may also have utility when Earth-derived fluvial sediment transport theory is applied to Mars and other extraterrestrial bodies (see Braat et al. 2024).

5 | Conclusions

Ever since Du Boys (1879) proposed that bedload transport was amenable to a theoretical treatment without observations, the overarching assumption has been that the rate of bedload transport is 'predictable'; such that a given flow of water acting on explicitly sized sediment particles, with unlimited availability, realises a precise estimate of bedload transport capacity. Putting the issue of a predictive formula's applicability aside, a major problem with this assumption is that it does not relate well to arbitrarily varying conditions, or circumstances in which the requisite input parameters are ill-defined (Gomez and Church 1989). Another is that, even in steady flows, bedload transport rates are known to be temporally variable (Gomez et al. 1989). Thus, predictions of bedload transport rates constitute a form of augmented reality, which portray the transport relation by combining a description of the real world (that typically relies on time-averaged input parameters) with empirical or theoretical elements (e.g., Du Boys 1879; Meyer-Peter and Müller 1948; Einstein 1950).

Nearly eight decades ago it was recognised that relations devised to fit experimental data obtained in laboratory flumes are not generally applicable, and their application to rivers involves an element of judgement and has to be tempered with caution (see Vanoni 1948; Anonymous 1954). Our analysis suggests that the trends revealed in data from prior experimental studies and comparable measurements made in rivers resemble, but do not duplicate one another. The separation arises because, except in rivers where there is good reason to suppose that particles exposed to the flow or already in motion help to determine the transport rate, rather than relying on a single value we use Equation (3)) to specify, based on the representative size (D_{50}) of the sediment in motion, the appropriate residual value of shear stress or stream power (via Equation (4)) required to maintain transport. Accepting that there is considerable uncertainty

surrounding the true value of the threshold shear stress that corresponds to a given sediment diameter in rivers, which is not a true material property and is difficult to specify (Buffington and Montgomery 1997), and recognising that Shields similarity can be preserved in geometrically distorted prototypes (see Einstein and Chien 1956), we contend that it is time to revisit the issue of the transferability of bedload transport formulae based almost entirely on laboratory data to the field.

The separation between laboratory and field data leads us to develop discrete relations, but the coherence of the field data, which encompass at-a-point samples collected in perennial, sand-bed rivers that are hundreds or thousands of meters wide and continuous measurements made, at or near the point motion begins, in ephemeral, gravel-bed rivers that are only a few meters wide, is extraordinary (Figure 3). In all these rivers there is near equivalence between the calibre of the bedload and bed material (Figure 2); the bed at the measurement site is fully mobile; there are no constraints on the availability of transportable material; and, we presume, the maximum amount of material that can be moved is indeed being moved. As an independent test confirms (Figure 7), we have confidence that our formulation (Equation (7); Table 1; Figure 5) is not only capable of capturing the overall trend observed under equilibrium conditions in natural rivers, across transport regimes that range from sporadic entrainment events on an otherwise quiescent bed, to intense and continual motion, but can also accommodate the inherently variable nature of the transport process. That said, as previously noted, because of the dependency ω has on U^3 , the relation that relies on ω^*/ω_o^* is necessarily more diffuse.

Finally, although there is frustration because 'it is still difficult to predict bedload transport rates more accurately than to within one order of magnitude' (Ancy 2020, 1), we contend that, as field data are fraught with variability and uncertainty, the requirement for any formula to exactly replicate in situ field measurements is a chimera. To this end we note that the range of variability in the field data we have assembled is similar to that seen in the individual records of concentration that routinely are used to develop rating curves for calculating river suspended sediment loads (see Warrick 2015). Consequently, it may be time to concede that variability is not an impediment to (and accept that it is an intrinsic property of populations and phenomena that assists) understanding.

Acknowledgements

This article is dedicated to the memory of G.P. (Gar) Williams, whose meticulous experiments demonstrated that the increase in bedload transport rates with decreasing flow depth is an inherent property of the transport process. B.G. is grateful to the many colleagues who have shared data over the years; we are also appreciative of all those researchers who have catalogued and made bedload transport data publicly available, and we thank the two reviewers for their encouraging comments about our work.

Data Availability Statement

The data that support the findings of this study are provided as [Supporting Information](#).

References

- Ancey, C. 2020. "Bedload Transport: A Walk Between Randomness and Determinism. Part 1. The State of the Art." *Journal of Hydraulic Research* 58, no. 1: 1–17. <https://doi.org/10.1080/00221686.2019.1702594>.
- Ancey, C., and A. Recking. 2023. "Scaling Behavior of Bedload Transport: What if Bagnold Was Right?" *Earth-Science Reviews* 246: 104571. <https://doi.org/10.1016/j.earscirev.2023.104571>.
- Anonymous. 1954. "Notes From the Fontana Conference on Sediment Transport." October 11–15, 1954, Fontana Village, North Carolina. 39 p. <https://catalog.hathitrust.org/Record/009219201>.
- Armoza-Zvuloni, R., Y. Shlomi, I. Abadi, R. Shem-Tov, and J. B. Laronne. 2022. "Fluvial Sediment Yields in Hyper-Arid Areas, Exemplified by Nahal Nehushtan, Israel." *Land* 11: 1050. <https://doi.org/10.3390/land11071050>.
- Badoux, A., N. Andres, and J. M. Turowski. 2014. "Damage Costs due to Bedload Transport Processes in Switzerland." *Natural Hazards and Earth System Sciences* 14, no. 2: 279–294. <https://doi.org/10.5194/nhess-14-279-2014>.
- Bagnold, R. A. 1960. "Sediment Discharge and Stream Power: A Preliminary Announcement." U.S. Geological Survey Circular 421, 23 p. <https://doi.org/10.3133/cir421>.
- Bagnold, R. A. 1966. "An Approach to the Sediment Transport Problem From General Physics." U.S. Geological Survey Professional Paper 422-I, U.S. Department of the Interior, Washington, D.C. 37 p. <https://doi.org/10.3133/pp422i>.
- Bagnold, R. A. 1977. "Bed Load Transport by Natural Rivers." *Water Resources Research* 13, no. 2: 303–312. <https://doi.org/10.1029/wr013i002p00303>.
- Bagnold, R. A. 1980. "An Empirical Correlation of Bedload Transport Rates in Flumes and Natural Rivers." *Proceedings of the Royal Society of London A. Mathematical and Physical Sciences* 372, no. 1751: 453–473. <https://doi.org/10.1098/rspa.1980.0122>.
- Bakke, P. D., L. S. Sklar, D. R. Dawdy, and W. C. Wang. 2017. "The Design of a Site-Calibrated Parker–Klingeman Gravel Transport Model." *Water* 9, no. 6: 441. <https://doi.org/10.3390/w9060441>.
- Braat, L., M. Z. M. Bruckner, E. Sefton-Nash, and M. P. Lamb. 2024. "Gravity-Driven Differences in Fluvial Sediment Transport on Mars and Earth." *Journal of Geophysical Research: Planets* 129: 7788. <https://doi.org/10.1029/2023JE007788>.
- Buffington, J. M., and D. R. Montgomery. 1997. "A Systematic Analysis of Eight Decades of Incipient Motion Studies, With Special Reference to Gravel-Bedded Rivers." *Water Resources Research* 33, no. 8: 1993–2029. <https://doi.org/10.1029/96wr03190>.
- Carson, M. A., and G. A. Griffiths. 1987. "Bedload Transport in Gravel Channels." *Journal of Hydrology New Zealand* 26, no. 1: 1–151.
- Chien, N. 1954. "Meyer-Peter Formula for Bed-Load Transport and Einstein Bed-Load Function." M.R.D. Sediment Series, No. 7, University of California, Institute of Engineering Research, Berkeley, California, 31 p.
- Cohen, H., J. B. Laronne, and I. Reid. 2010. "Simplicity and Complexity of Bed Load Response During Flash Floods in a Gravel Bed Ephemeral River: A 10 Year Field Study." *Water Resources Research* 46: W11542. <https://doi.org/10.1029/2010wr009160>.
- Dey, S., and S. Z. Ali. 2019. "Bed Sediment Entrainment by Streamflow: State of the Science." *Sedimentology* 66: 1449–1485. <https://doi.org/10.1111/sed.12566>.
- Dinehart, R. L. 1992. "Evolution of Coarse Gravel Bed Forms: Field Measurements at Flood Stage." *Water Resources Research* 28, no. W11542: 2667–2689. <https://doi.org/10.1029/92wr01357>.
- Du Boys, P. 1879. "Le Rhône et les rivières à lit affouillable. Etude du régime du Rhône et de l'action exercée par les eaux sur un lit à fond de graviers indéfiniment affouillable." *Annales des Ponts et Chaussées* 49, no. 2: 141–195. <https://gallica.bnf.fr/ark:/12148/bpt6k408532b/f139.item#>.
- Eaton, B. C., and M. Church. 2011. "A Rational Sediment Transport Scaling Relation Based on Dimensionless Stream Power." *Earth Surface Processes and Landforms* 36, no. 7: 901–910. <https://doi.org/10.1002/esp.2120>.
- Einstein, H. A. 1942a. "Formulas for the Transportation of Bed Load." *Transactions of the American Society of Civil Engineers* 107, no. 1: 561–573. <https://doi.org/10.1061/taceat.0005468>.
- Einstein, H. A. 1942b. "Flow on a Movable Bed." Proceedings of the 2nd Hydraulics Conference University of Iowa, 332–341.
- Einstein, H. A. 1944. "Bed-Load Transportation in Mountain Creek." In *Greenville Sediment Load Laboratory SCS-TP- 55*, 54. U.S. Department of Agriculture Soil Conservation Service.
- Einstein, H. A. 1950. "The Bedload Function for Sediment Transportation in Open Channels." Technical Bulletin 1026, U.S. Department of Agriculture Soil Conservation Service, Washington, D.C., 70 p.
- Einstein, H. A., and N. Chien. 1956. "Similarity of Distorted River Models With Movable Beds." *Transactions of the American Society of Civil Engineers* 121, no. 1: 440–457. <https://doi.org/10.1061/taceat.0007300>.
- Fenton, J. D., and J. E. Abbott. 1977. "Initial Movement of Grains on a Stream Bed: The Effect of Relative Protrusion." *Proceedings of the Royal Society of London. A. Mathematical and Physical Sciences* 352, no. 1671: 523–537. <https://doi.org/10.1098/rspa.1977.0014>.
- Ferguson, R. I. 2005. "Estimating Critical Stream Power for Bedload Transport Calculations in Gravel-Bed Rivers." *Geomorphology* 70, no. 1–2: 33–41. <https://doi.org/10.1016/j.geomorph.2005.03.009>.
- Gelman, A. 2017. "A Continuous Hinge Function for Statistical Modeling." <https://statmodeling.stat.columbia.edu/2017/05/19/>.
- Gilbert, G. K. 1876. "The Colorado Plateau and Provinces as a Field for Geological Study." *American Journal of Science and Arts* 12: 16–24. <https://doi.org/10.2475/ajs.s3-12.67.16>.
- Gilbert, G. K. 1914. "The Transportation of Debris by Running Water." U.S. Geological Survey Professional Paper 86, U.S. Department of the Interior, Washington, D.C. 263 p.
- Gomez, B. 2006. "The Potential Rate of Bed-Load Transport." *Proceedings of the National Academy of Sciences* 103, no. 46: 17170–17173. <https://doi.org/10.1073/pnas.0608487103>.
- Gomez, B. 2022. "The Efficiency of the River Machine." *Geomorphology* 410: 108271. <https://doi.org/10.1016/j.geomorph.2022.108271>.
- Gomez, B., and M. Church. 1988. "A Catalogue of Equilibrium Bedload Transport Data for Coarse Sand and Gravel-Bed Channels." Department of Geography, University of British Columbia, Vancouver. 90 p.
- Gomez, B., and M. Church. 1989. "An Assessment of Bed Load Sediment Transport Formulae for Gravel Bed Rivers." *Water Resources Research* 25, no. 6: 1161–1186. <https://doi.org/10.1029/wr025i006p01161>.
- Gomez, B., R. L. Naff, and D. W. Hubbell. 1989. "Temporal Variations in Bedload Transport Rates Associated With the Migration of Bedforms." *Earth Surface Processes and Landforms* 14, no. 2: 135–156. <https://doi.org/10.1002/esp.3290140205>.
- Gomez, B., and P. J. Soar. 2022. "Bedload Transport: Beyond Intractability." *Royal Society Open Science* 9, no. 3: 211932. <https://doi.org/10.1098/rsos.211932>.
- Gomez, B., and P. J. Soar. 2023. "Bedload Transport and the Stream Power Approach." *Proceedings of the Royal Society of London. Series A:*

- Mathematical and Physical Sciences* 479, no. 2273: 20220783. <https://doi.org/10.1098/rspa.2022.0783>.
- Gomez, B., P. J. Soar, and P. W. Downs. 2022. "Good Vibrations: Big Data Impact Bedload Research." *Earth Surface Processes and Landforms* 47, no. 1: 129–142. <https://doi.org/10.1002/ESP.5304>.
- Hayes, S. K., D. R. Montgomery, and C. G. Newhall. 2002. "Fluvial Sediment Transport and Deposition Following the 1991 Eruption of Mount Pinatubo." *Geomorphology* 45, no. 3–4: 211–224. [https://doi.org/10.1016/s0169-555x\(01\)00155-6](https://doi.org/10.1016/s0169-555x(01)00155-6).
- Hinton, D., R. Hotchkiss, and D. P. Ames. 2017. "Comprehensive and Quality-Controlled Bedload Transport Database." *Journal of Hydraulic Engineering* 143, no. 2: 6016024. [https://doi.org/10.1061/\(asce\)hy.1943-7900.0001221](https://doi.org/10.1061/(asce)hy.1943-7900.0001221).
- Hudson, H. R. 1983. "Discussion of Bedload and Size Distribution in Paved Gravel-Bed Streams by Gary Parker, Peter C. Klingeman, and David G. McLean (April, 1982)." *Journal of Hydraulic Engineering* 109, no. 5: 789–791. [https://doi.org/10.1061/\(asce\)0733-9429\(1983\)109:5\(789.2\)](https://doi.org/10.1061/(asce)0733-9429(1983)109:5(789.2)).
- Ikeda, H. 1983. "Experiments on Bedload Transport, Bed Forms, and Sedimentary Structures Using Fine Gravel in the 4-Meter-Wide Flume." Environmental Research Center Paper 2, The University of Tsukuba, Tsukuba, Ibaraki, Japan, 78 p.
- Inbar, M., and A. P. Schick. 1979. "Bedload Transport Associated With High Stream Power, Jordan River, Israel." *Proceedings of the National Academy of Sciences* 76, no. 6: 2515–2517. <https://doi.org/10.1073/pnas.76.6.2515>.
- Johnson, J. W. 1939. "Discussion of Laboratory Investigations of Flume Traction and Transportation by Y.L. Chang." *Transactions of the American Society of Civil Engineers* 104, no. 1: 1247–1313.
- Knott, J. M., S. W. Lipscomb, and T. W. Lewis. 1987. "Sediment Transport Characteristics of Selected Streams in the Susitna River Basin, Alaska; Data for Water Year 1985 and Trends in Bedload Discharge, 1981–85." U.S. Geological Survey Open-File Report 87–229, 54 p. <https://doi.org/10.3133/ofr87229>.
- Kuhnle, R. A. 1993. "Incipient Motion of Sand-Gravel Sediment Mixtures." *Journal of Hydraulic Engineering* 119, no. 12: 1400–1415. [https://doi.org/10.1061/\(asce\)0733-9429\(1993\)119:12\(1400\)](https://doi.org/10.1061/(asce)0733-9429(1993)119:12(1400)).
- Lane, S. N., G. Antoniazza, M. Dietze, and D. Mancini. 2021. "Bedload Transport: Are We Doing Restoration Right in the Face of Alpine Climate Change?" In *Proceedings of the International Symposium on Bedload Management 2021, 8–10 November 2021*, edited by C. R. Wyss, G. de Cesare, S. N. Lane, et al., 60–73. Wasser-Agenda 21. <https://doi.org/10.3929/ethz-b-000513098>.
- Lenzi, M. A. 2001. "Step–Pool Evolution in the Rio Cordon, Northeastern Italy." *Earth Surface Processes and Landforms* 26, no. 9: 991–1008. <https://doi.org/10.1002/esp.239>.
- Leopold, L. B., and W. W. Emmett. 1976. "Bedload Measurements, East Fork River, Wyoming." *Proceedings of the National Academy of Sciences* 73, no. 4: 1000–1004. <https://doi.org/10.1073/pnas.73.4.1000>.
- Leopold, L. B., and W. W. Emmett. 1997. *Bedload and River Hydraulics: Inferences From the East Fork River, Wyoming*, 52. U.S. Geological Survey Professional Paper 1583, U.S. Department of the Interior. <https://doi.org/10.3133/pp1583>.
- Meade, R. H. 1985. "Wavelike Movement of Bedload Sediment, East Fork River, Wyoming." *Environmental Geology and Water Sciences* 7, no. 4: 215–225. <https://doi.org/10.1007/bf02509922>.
- Meyer-Peter, E., and R. Müller. 1948. "Formulas for Bed-Load Transport." In *Proceedings of the 2nd Meeting, International Association for Hydraulic Structures Research*, vol. 2, 39–64. International Association for Hydraulic Research, Madrid, Appendix.
- Milhouse, R. T. 1973. "Sediment Transport in a Gravel-Bottomed Stream." Unpublished Ph.D. diss., Oregon State University, Corvallis, OR. 231 p.
- Monsalve, A. D., E. M. Yager, J. M. Turowski, and D. Rickenmann. 2016. "A Probabilistic Formulation of Bed Load Transport to Include Spatial Variability of Flow and Surface Grain Size Distributions." *Water Resources Research* 52, no. 5: 3579–3598. <https://doi.org/10.1002/2015WR017694>.
- Montgomery, D. R., and J. M. Buffington. 1997. "Channel-Reach Morphology in Mountain Drainage Basins." *Geological Society of America Bulletin* 109, no. 5: 596–611. [https://doi.org/10.1130/0016-7606\(1997\)109<0596:crmimd>2.3.co;2](https://doi.org/10.1130/0016-7606(1997)109<0596:crmimd>2.3.co;2).
- Montgomery, D. R., M. S. Panfil, and S. K. Hayes. 1999. "Channel-Bed Mobility Response to Extreme Sediment Loading at Mount Pinatubo." *Geology* 27, no. 3: 271–274. [https://doi.org/10.1130/0091-7613\(1999\)027%3c0271:cbmrte%3e2.3.co;2](https://doi.org/10.1130/0091-7613(1999)027%3c0271:cbmrte%3e2.3.co;2).
- Neill, C. R., J. S. Buska, E. F. Chacho, C. M. Collins, and L. W. Gatto. 1984. "Overview of Tanana River Monitoring and Research Studies Near Fairbanks, Alaska." U.S. Army Cold Regions Research and Engineering Laboratory, Hanover, N.H., 360 p.
- Paintal, A. S. 1971. "Concept of Critical Shear Stress in Loose Boundary Open Channels." *Journal of Hydraulic Research* 9, no. 1: 91–113. <https://doi.org/10.1080/00221687109500339>.
- Paphitis, D. 2001. "Sediment Movement Under Unidirectional Flows: An Assessment of Empirical Threshold Curves." *Coastal Engineering* 43, no. 3–4: 227–245. [https://doi.org/10.1016/S0378-3839\(01\)00015-1](https://doi.org/10.1016/S0378-3839(01)00015-1).
- Parker, G. 1990. "Surface-Based Bedload Transport Relation for Gravel Rivers." *Journal of Hydraulic Research* 28, no. 4: 417–436. <https://doi.org/10.1080/00221689009499058>.
- Picco, L., L. Mao, E. Rignon, et al. 2012. "An Update of the Sediment Fluxes Investigation in the Rio Cordon (Italy) After 25 Years of Monitoring." *Journal of Agricultural Engineering* 43, no. 3: 108–113. <https://doi.org/10.4081/jae.2012.e17>.
- Piñeiro, G., S. Perelman, J. P. Guerschman, and J. M. Paruelo. 2008. "How to Evaluate Models: Observed vs. Predicted or Predicted vs. Observed?" *Ecological Modelling* 216, no. 3–4: 316–322. <https://doi.org/10.1016/j.ecolmodel.2008.05.006>.
- Proffitt, G. T., and A. J. Sutherland. 1983. "Transport of Non-Uniform Sediments." *Journal of Hydraulic Research* 21, no. 1: 33–43. <https://doi.org/10.1080/00221688309499448>.
- Recking, A. 2006. "An Experimental Study of Grain Sorting Effects on Bedload." Unpublished Ph.D. diss., Institut National des Sciences Appliquées de Lyon, Lyon, France. 257 p.
- Recking, A. 2010. "A Comparison Between Flume and Field Bed Load Transport Data and Consequences for Surface-Based Bed Load Transport Prediction." *Water Resources Research* 46, no. 3: W03518. <https://doi.org/10.1029/2009wr008007>.
- Reid, I., and J. B. Laronne. 1995. "Bed Load Sediment Transport in an Ephemeral Stream and a Comparison With Seasonal and Perennial Counterparts." *Water Resources Research* 31, no. 3: 773–781. <https://doi.org/10.1029/94wr02233>.
- Samaga, B. R., K. G. R. Raju, and R. J. Garde. 1986. "Bed Load Transport of Sediment Mixtures." *Journal of Hydraulic Engineering* 112, no. 11: 1003–1017. [https://doi.org/10.1061/\(asce\)0733-9429\(1986\)112:11\(1003\)](https://doi.org/10.1061/(asce)0733-9429(1986)112:11(1003)).
- Shih, W. R., and P. Diplas. 2018. "A Unified Approach to Bed Load Transport Description Over a Wide Range of Flow Conditions via the Use of Conditional Data Treatment." *Water Resources Research* 54, no. 5: 3490–3509. <https://doi.org/10.1029/2017WR022373>.
- Simon, A. 1999. "Channel and Drainage Basin Response of the Toutle River System in the Aftermath of the 1980 Eruption of Mount St. Helens, Washington." U.S. Geological Survey Open-File Report. 96-633, 130 p. <https://doi.org/10.3133/ofr96633>.

Smart, G. M. 1984. "Sediment Transport Formula for Steep Channels." *Journal of Hydraulic Engineering* 110, no. 3: 267–276. [https://doi.org/10.1061/\(asce\)0733-9429\(1984\)110:3\(267\)](https://doi.org/10.1061/(asce)0733-9429(1984)110:3(267)).

Soulsby, R. L., and R. J. Whitehouse. 1997. "Threshold of Sediment Motion in Coastal Environments." Proceedings of the 13th Australasian Coastal and Ocean Engineering Conference and the 6th Australasian Port and Harbour Conference, Christchurch, N.Z. Centre for Advanced Engineering, University of Canterbury, Christchurch, N.Z. 1, 145–150.

Urich, M. A., K. R. Spicer, A. R. Mosbrucker, D. R. Saunders, and T. S. Christianson. 2021. "A 40-Year Story of River Sediment at Mount St. Helens." U.S. Geological Survey Fact Sheet 2021–3004. <https://doi.org/10.3133/fs20213004>.

USACE. 2023. "HEC-RAS 2D Sediment Technical Reference Manual." U.S. Army Corps of Engineers, Hydrologic Engineering Centre, Davis, CA. 101 p. <https://www.hec.usace.army.mil/confluence/rasdocs/d2sd/ras2dsedtr/latest>.

Vanoni, V. A. 1948. "Development of the Mechanics of Sediment Transportation." Proceedings of the Federal Inter-Agency Sedimentation Conference, Denver, Colo., May 6–8, 1947, Bureau of Reclamation, U.S. Dept Interior, Washington, D.C., 209–224.

Vanoni, V. A., and N. H. Brooks. 1957. "Laboratory Studies of the Roughness and Suspended Load of Alluvial Streams." Missouri River Division Sediment Series 11, Sedimentation Laboratory, California Institute of Technology, Pasadena, California, U.S.A., 121 p.

Vanoni, V. A., N. H. Brooks, and J. F. Kennedy. 1961. "Lecture Notes on Sediment Transportation and Channel Stability." W.M. Keck Laboratory of Hydraulics and Water Resources, California Institute of Technology, Pasadena, CA, 150 p.

Warburton, J. 1992. "Observations of Bed Load Transport and Channel Bed Changes in a Proglacial Mountain Stream." *Arctic and Alpine Research* 24, no. 3: 195–203. <https://doi.org/10.2307/1551657>.

Warrick, J. A. 2015. "Trend Analyses With River Sediment Rating Curves." *Hydrological Processes* 29, no. 6: 936–949. <https://doi.org/10.1002/hyp.10198>.

Wilcock, P. R. 2001. "Toward a Practical Method for Estimating Sediment-Transport Rates in Gravel-Bed Rivers." *Earth Surface Processes and Landforms* 26, no. 13: 1395–1408. <https://doi.org/10.1002/esp.301>.

Wilcock, P. R., and J. C. Crowe. 2003. "Surface-Based Transport Model for Mixed-Size Sediment." *Journal of Hydraulic Engineering* 129, no. 2: 120–128. [https://doi.org/10.1061/\(ASCE\)0733-9429\(2003\)129:2\(120\)](https://doi.org/10.1061/(ASCE)0733-9429(2003)129:2(120)).

Wilcock, P. R., and B. T. DeTemple. 2005. "Persistence of Armor Layers in Gravel-Bed Streams." *Geophysical Research Letters* 32, no. 8: L08402. <https://doi.org/10.1029/2004GL021772>.

Wilcock, P. R., and J. B. Southard. 1989. "Bed Load Transport of Mixed Size Sediment: Fractional Transport Rates, Bed Forms, and the Development of a Coarse Bed Surface Layer." *Water Resources Research* 25, no. 7: 1629–1641. <https://doi.org/10.1029/wr025i007p01629>.

Williams, G. P. 1970. "Flume Width and Water Depth Effects in Sediment-Transport Experiments." U.S. Geological Survey Professional Paper 562-H, U.S. Department of the Interior, Washington, D.C. 37 p. <https://doi.org/10.3133/pp562h>.

Wong, M., and G. Parker. 2006. "Reanalysis and Correction of Bed-Load Relation of Meyer-Peter and Müller Using Their Own Database." *Journal of Hydraulic Engineering* 132, no. 11: 1159–1168. <https://doi.org/10.1029/2006WR005172>.

Yuill, B., M. Nichols, and E. Yager. 2010. "Coarse Bed Material Patch Evolution in Low-Order, Ephemeral Channels." *Catena* 81, no. 2: 126–136. <https://doi.org/10.1016/j.catena.2010.02.002>.

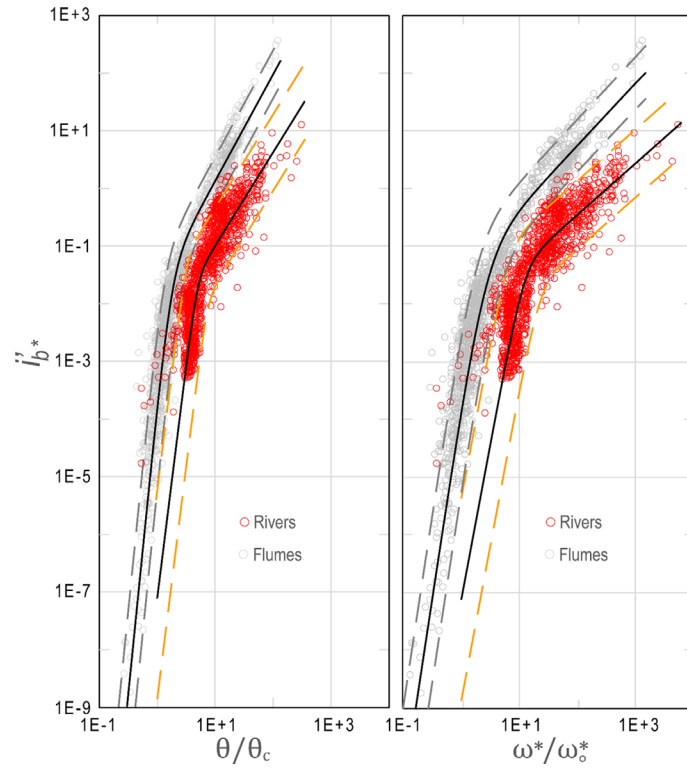
Yuill, B. T. 2009. "Sediment Transport and Bed Mobility in a Low-Ordered Ephemeral Watershed." Unpublished Ph.D. diss., The University of Arizona, Tucson, AZ. 199 p.

Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Graphical Abstract

Please note that Graphical Abstracts only appear online as part of a table of contents and are not part of the main article (therefore, they do not appear in the article HTML or PDF files).



Diverse field and laboratory measurements of equilibrium bedload transport collapse onto single curves that mirror, but do not duplicate one another, and can be characterised by integrated relations that permit the transport rate to be computed for conditions that range from sporadic entrainment events on an otherwise quiescent bed in small ephemeral rivers to intense and continual motion in large perennial rivers.